

APIA FALEOLO, Samoa

WMO: 917590

Lat: 13.830S

Lon: 172.008W

Elev: 18

StdP: 101.11

Time Zone: 13.00 (PCS)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	21.8	18.7	13.6	23.9	19.3	14.1	23.9	10.7	29.4	9.6	29.1	1.1	180	0.343

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.2	32.2	26.5	32.1	26.4	31.8	26.3	27.6	30.8	27.2	30.6	27.0	30.4	5.7	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.8	22.4	29.8	26.2	21.7	29.3	26.1	21.5	29.2	87.8	31.1	86.4	31.0	85.3	30.5	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.9	7.7	DB	19.3	33.6	1.3	0.5	18.4	33.9	17.7	34.2	16.9	34.5	16.0	34.9	(4)
(5)			WB	18.1	28.9	1.6	0.9	17.0	29.5	16.1	30.0	15.2	30.5	14.0	31.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.1	27.6	27.7	27.7	27.6	27.2	26.9	26.4	26.7	26.9	27.2	27.4	(6)	
(7)		DBStd	0.99	0.97	0.91	0.82	0.77	0.86	0.86	0.98	0.87	0.87	0.84	0.82	0.89	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6249	546	495	548	528	533	507	507	507	501	523	517	538	(10)
(11)		CDD18.3	3207	287	262	289	278	274	257	249	249	251	265	267	280	(11)
(12)		CDH23.3	30825	2973	2721	2981	2771	2586	2319	2117	2150	2314	2501	2602	2790	(12)
(13)		CDH26.7	10181	976	906	1024	958	849	751	658	691	786	834	849	899	(13)
(14)	Wind	WSAvg	2.6	3.2	2.9	2.5	2.2	2.1	2.5	2.6	2.8	3.0	2.7	2.4	2.5	(14)
(15)	Precipitation	PrecAvg	2115	300	249	209	169	168	90	95	83	114	170	221	246	(15)
(16)		PrecMax	2516	629	606	409	535	377	215	218	201	274	312	398	378	(16)
(17)		PrecMin	1467	122	57	48	39	14	34	29	12	16	70	112	107	(17)
(18)		PrecStd	225	124	133	95	104	79	39	48	44	56	55	72	64	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.7	32.8	32.9	32.9	32.2	32.2	32.1	32.1	32.3	32.2	32.2	32.3	(19)
(20)			MCWB	27.0	26.9	27.0	26.9	26.7	26.0	25.7	25.8	25.9	26.1	26.3	26.8	(20)
(21)		2%	DB	31.9	31.9	32.0	32.0	31.8	31.2	31.2	31.2	31.7	31.8	31.8	31.7	(21)
(22)			MCWB	26.7	26.6	26.7	26.6	26.5	25.9	25.5	25.5	25.7	26.0	26.3	26.5	(22)
(23)		5%	DB	31.1	31.1	31.2	31.2	31.1	30.9	30.7	30.7	31.0	31.0	31.0	31.0	(23)
(24)			MCWB	26.6	26.5	26.5	26.5	26.4	25.9	25.4	25.4	25.5	25.8	26.1	26.3	(24)
(25)		10%	DB	30.7	30.8	30.9	30.9	30.5	30.2	30.0	30.0	30.2	30.2	30.2	30.3	(25)
(26)			MCWB	26.4	26.4	26.4	26.4	26.1	25.5	25.2	25.1	25.2	25.5	25.8	25.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.7	27.7	27.7	27.7	27.8	27.2	27.1	26.7	27.2	27.5	27.4	27.7	(27)
(28)			MCDWB	30.7	30.6	31.1	31.0	30.7	30.5	30.7	30.5	30.7	30.9	30.9	31.1	(28)
(29)		2%	WB	27.2	27.2	27.2	27.2	27.0	26.7	26.5	26.2	26.5	26.7	26.7	27.0	(29)
(30)			MCDWB	30.4	30.4	30.6	30.7	30.6	30.2	30.0	29.9	30.2	30.3	30.3	30.4	(30)
(31)		5%	WB	26.9	26.8	26.9	26.7	26.6	26.3	26.0	25.8	26.0	26.2	26.4	26.6	(31)
(32)			MCDWB	30.1	30.0	30.2	30.2	30.1	29.8	29.4	29.4	29.8	29.8	29.9	29.9	(32)
(33)		10%	WB	26.6	26.5	26.5	26.5	26.3	25.9	25.6	25.4	25.6	25.8	26.1	26.2	(33)
(34)			MCDWB	29.6	29.6	29.8	29.8	29.6	29.3	29.0	29.0	29.3	29.3	29.5	29.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.1	6.2	6.7	7.0	7.1	7.3	7.5	7.7	7.6	7.3	6.8	6.5	(35)
(36)			MCDBR	7.0	6.9	7.4	7.6	7.7	8.0	8.5	8.6	8.5	8.3	7.6	7.3	(36)
(37)			MCWBR	3.2	3.2	3.5	3.6	3.8	3.9	4.2	4.2	4.1	4.1	3.7	3.5	(37)
(38)		5% WB	MCDBR	6.4	6.5	6.9	7.1	7.2	7.4	7.6	7.8	7.8	7.4	7.0	6.7	(38)
(39)			MCWBR	3.1	3.2	3.4	3.6	3.7	3.9	4.1	4.1	4.1	4.0	3.6	3.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.386	0.378	0.375	0.382	0.391	0.389	0.391	0.394	0.397	0.387	0.385	0.385	(40)	
(41)		taud	2.626	2.658	2.668	2.641	2.597	2.588	2.576	2.558	2.552	2.595	2.614	2.623	(41)	
(42)		Ebn at Noon	957	957	941	902	861	845	852	877	906	937	952	957	(42)	
(43)		Edh at Noon	102	98	94	92	91	89	92	98	104	103	102	102	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.71	5.69	5.71	5.14	4.54	4.33	4.48	5.19	5.81	6.00	5.90	5.81	(44)	
(45)		RadStd	0.56	0.54	0.41	0.41	0.31	0.22	0.18	0.24	0.32	0.30	0.31	0.36	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page